

Question – 01

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sharjeel < 04 >: run main
Vulnerable Towers:
A B E C

Communication Routes:
A B D
A B E H
A B E I
A C F
A C G

Longest Communication Chain:
A B E H

Level with Maximum Towers = 2 (4 towers)

Tower Failed: E
Isolated Communication Region:
E H I
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Question – 02

1. Determine which diseases are visited using Preorder.

Answer: Preorder Traversal (Root → Left → Right)

Sequence: Diagnosis, Infection, Viral, Bacterial, Deficiency, Iron, Vitamin

2. Determine which diseases are visited using Inorder.

Answer: Inorder Traversal (Left → Root → Right)

Sequence: Viral, Infection, Bacterial, Diagnosis, Iron, Deficiency, Vitamin

3. Determine which diseases are visited using Postorder.

Answer: Postorder Traversal (Left → Right → Root)

Sequence: Viral, Bacterial, Infection, Iron, Vitamin, Deficiency, Diagnosis

4. Determine which traversal should be used if the diagnosis engine must evaluate parent decisions before child decisions.

Answer: Preorder Traversal

Preorder naturally processes the current node (the parent) before expanding into its left and right descendants (the children).

5. Determine which traversal should be used if all child diagnoses must be completed before a final decision.

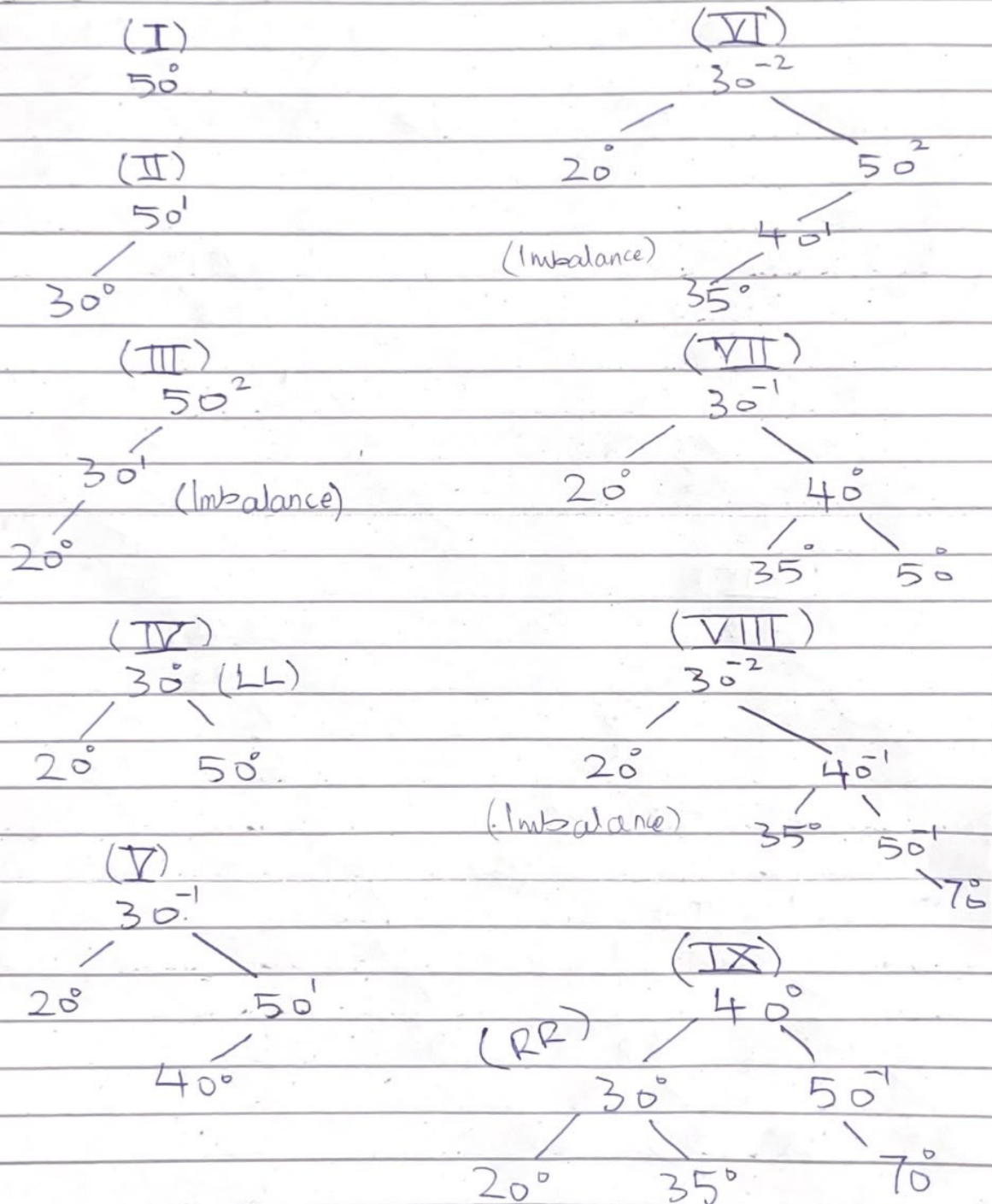
Answer: Postorder Traversal

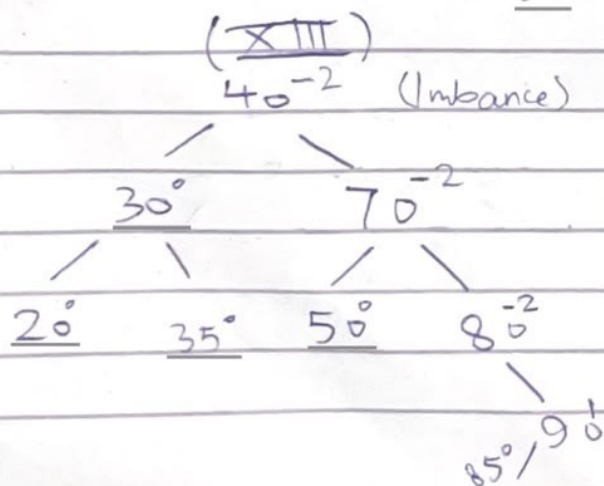
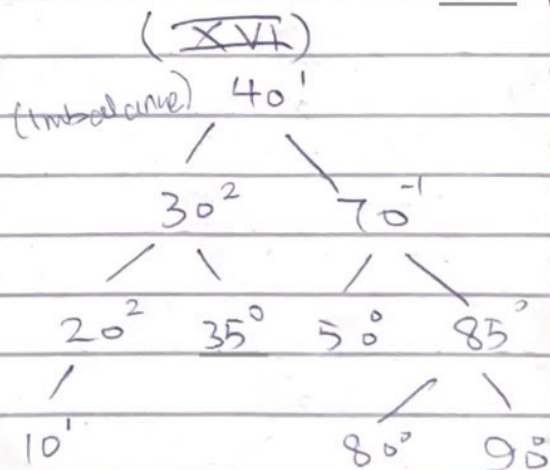
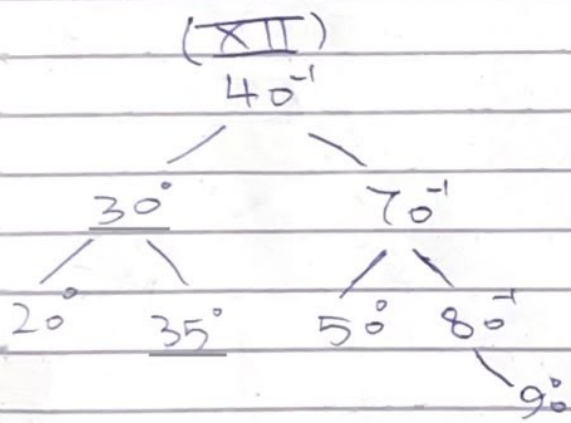
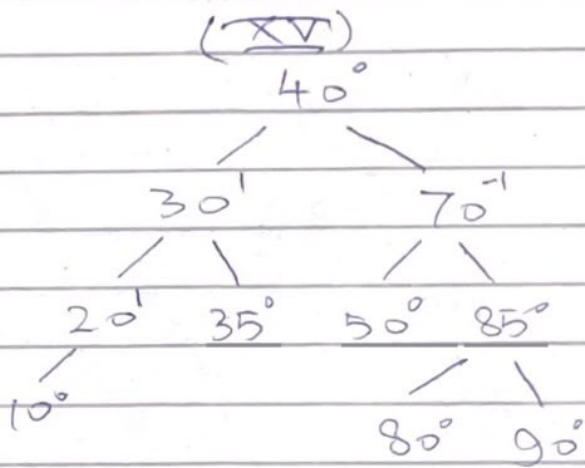
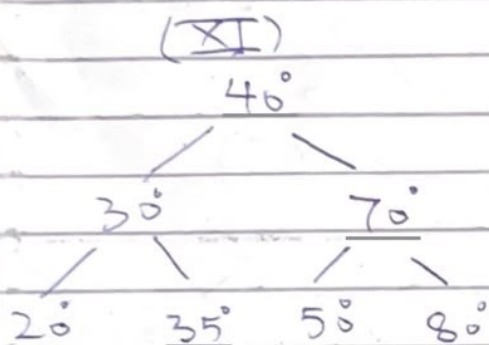
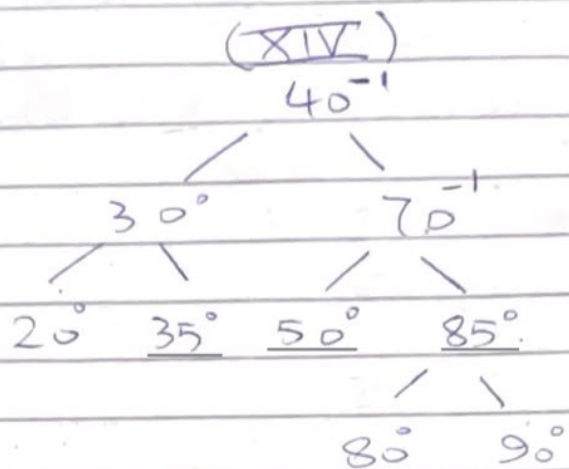
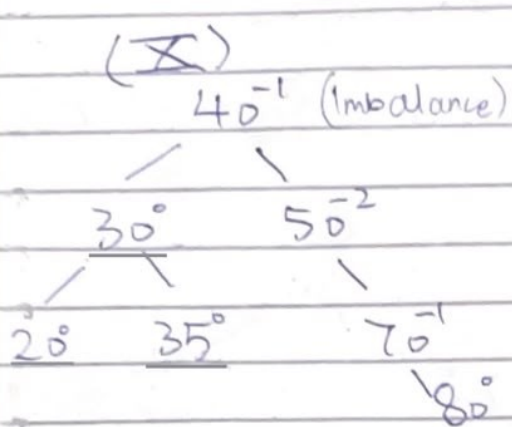
Postorder delays processing the parent node until both its left and right subtrees (the children) have been completely evaluated

Question - 03

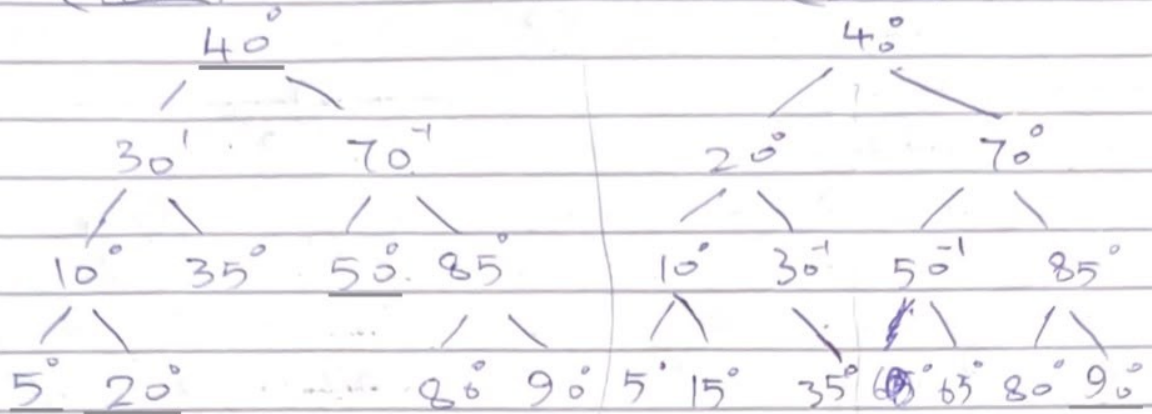
Question-03

50, 30, 20, 40, 35, 70, 80, 90, 85, 10, 5, 15, 65, 75, 72

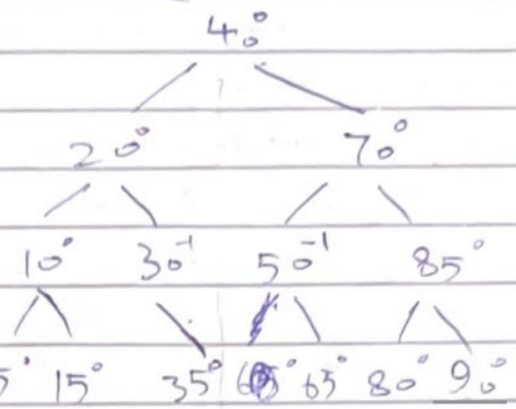




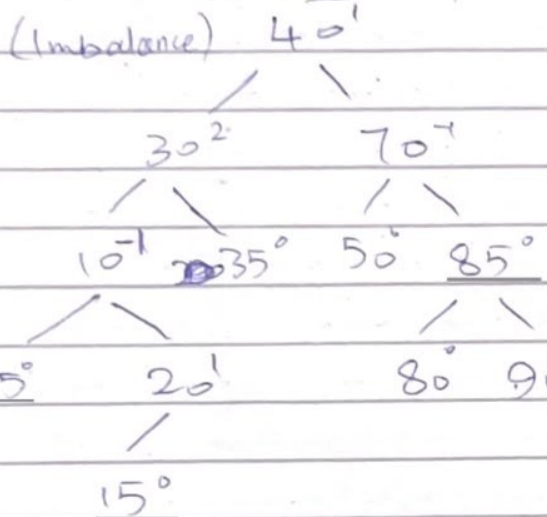
(XVII)



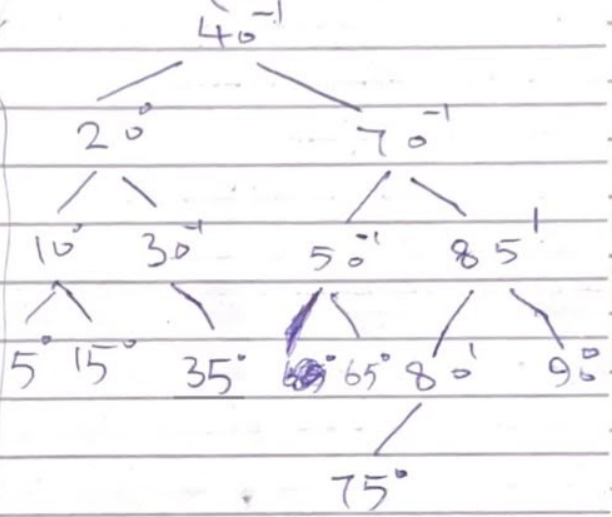
(XX)



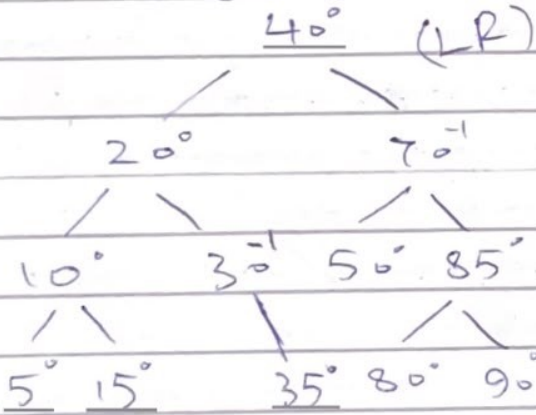
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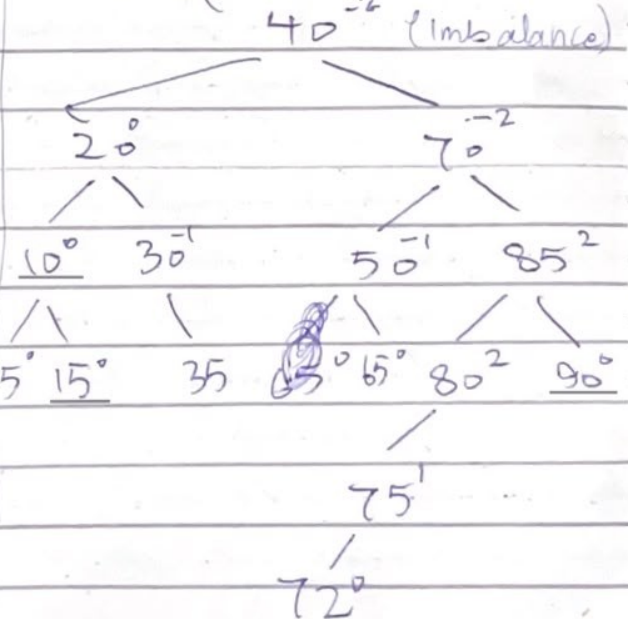
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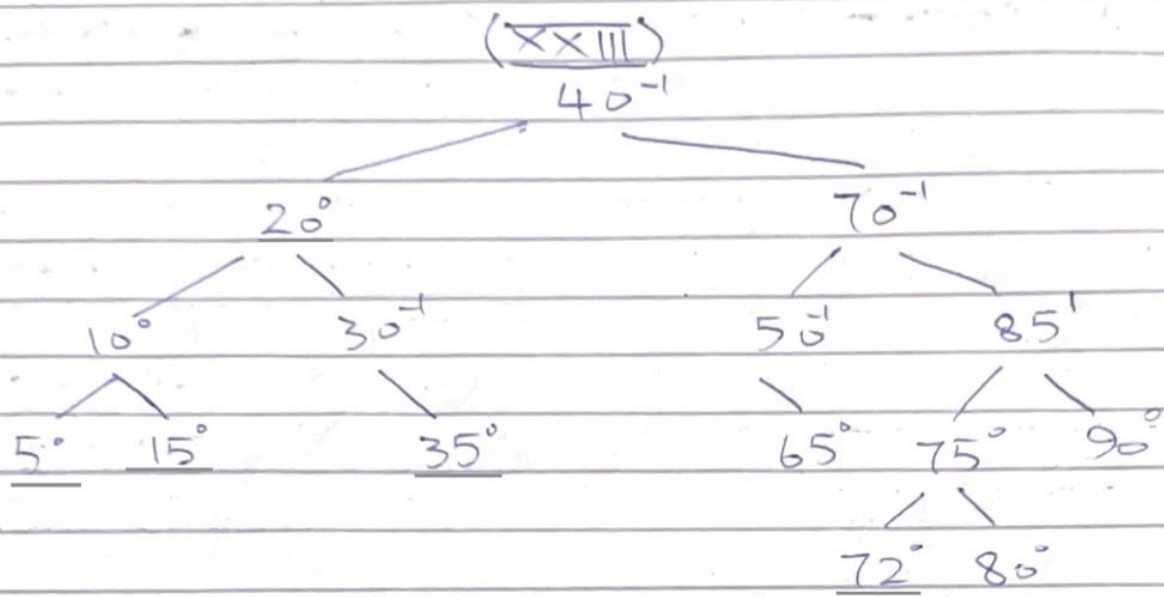


(XIX)



(XXII)





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